

Dear Students,

This study material serves as the threshold of your preparation. It is recommended that you should formulate figurative questions based on the provided definitions and derivations.

KMC Note

PERIODIC MOTION

A) DEFINITION

- I. Periodic motion (examples)
- II. Simple harmonic motion (examples) and its necessary conditions
- III. Characteristics of SHM (amplitude, displacement, velocity, acceleration) (with mathematical relation)
- IV. Simple pendulum and Seconds pendulum
- V. Free, Forced, damped and undamped oscillation, resonance

B) DERIVATION

- I. Total energy of SHM (with nature of graph)
- II. Motion of simple pendulum is simple harmonic and its time period
- III. Horizontal mass spring (calculation of time period)
- IV. Vertical mass spring (calculation of time period)

ROTATIONAL DYNAMICS

A) DEFINITION

- I. Analogues terms of linear motion with angular motion
- II. Moment of Inertia (formula and physical significance), Radius of gyration
- III. Torque, Angular momentum, Couple

B) DERIVATION

- I. Find moment of Inertia of thin uniform rigid rod about an axis through center and at one end.
- II. Principle of conservation of angular momentum and its verification with examples.
- III. Relation between torque and moment of inertia i.e. $\tau = I \alpha$
- IV. Relation between angular momentum and moment of inertia i.e. $L = I \omega$
- V. Relation between angular momentum and torque i.e. $\tau = dL/dt$
- VI. Work done by couple ($W = \tau \Theta$) and Power in rotational motion ($P = \tau \omega$)
- VII. Kinetic energy of Rotating body ($K.E = \frac{1}{2} I \omega^2$)

WAVE AND OPTICS

Section 1: WAVE MOTION:

KMC Note

A) DEFINITION

- I. Terms related to wave motion (frequency, wavelength, wave velocity, particle velocity, phase and Phase difference)
- II. Transverse wave and longitudinal wave.
- III. Progressive wave, stationary wave and nodes and antinodes.
- IV. Superposition principle.

B) DERIVATION

- I. Progressive wave equation.
- II. Stationary wave equation.

Section 2: MECHANICAL WAVE:

A) DEFINITION

- a) Newton's assumption and its derivation.
- b) Laplace's assumption and its derivation.
- c) Factor's affecting the velocity of sound in air.
 - i. Temperature
 - ii. Pressure
 - iii. Density
 - iv. Humidity or moisture

Section 3: Pipes and strings

A) DEFINITION

- I. Organ pipe
- II. Open and closed organ pipe
- III. Fundamental frequency, Harmonics, Overtones
- IV. End correction
(And Formula only: for open and closed pipe)

B) DERIVATION

- i. Modes of vibration of open organ pipe
- ii. Modes of vibration of closed organ pipe.
- iii. Modes of vibration in a stretched string.
- iv. Laws of transverse vibration of stretched string and its verification

MAGNETISM / MAGNETIC FIELD

KMC Note

A) DEFINITION

A) DERIVATION:

A) DEFINITION:

- i. Lorentz force: formula and Flemming's left-hand rule to determine the direction of magnetic Lorentz force
- ii. Moving coil galvanometer and current and voltage sensitivity.
- iii. Hall effect and hall probe (statement)
- iv. Formula for magnetic field due to a moving charge.
- v. Biot-Savart law (statement)
- vi. Ampere's circuital law (statement and proof)
- vii. Definition of 1 ampere current

A) DERIVATION:

- i. Torque experienced by a rectangular coil carrying current placed in the magnetic field.
- ii. Derivation of hall voltage and coefficient
- iii. Applications of Biot-Savart law: field on the center of circular coil, on the axis of circular coil, inside solenoid, straight conductor
- iv. Applications of Ampere's law: field due to solenoid, straight conductor, and toroid
- v. Force between current carrying parallel conductor and definition of one ampere current on the basis of force between current carrying parallel conductors.
- vi. Magnetic field of a moving charge
- vii. Force on a current carrying conductor by magnetic field

MODERN PHYSICS

ELECTRON:

KMC Note

A) DEFINITION

- i. Cathode Ray
- ii. Cross field.
- iii. Discharge

B) DERIVATION

- i. Motion of an electron in an electric field.
- ii. Motion of an electron in a magnetic field.
- iii. Millikan's Oil Drop Experiment (Quantization of charge).
- iv. J.J Thomson experiment (Specific charge of electron).

PHOTONS:

A) DEFINITION

- i. Photoelectric effect.
- ii. Threshold frequency
- iii. Threshold wavelength
- iv. Work function

B) DERIVATION

- i. Photoelectric equation.
- ii. Millikan's experiment to determine Planck's constant (with graph)

QUANTIZATION OF ENERGY:

A) DEFINITION

- i. Bohr's Postulate.
- ii. Excitation and Ionization Potential.
- iii. De Broglie Theory.
- iv. Heisenberg's Uncertainty Principle.

B) DERIVATION

- i. Bohr's theory of Hydrogen like atoms (Radius of the n-th orbits)
- iii. Energy of an electron in n-th orbits
- iii. Spectral Series of Hydrogen atoms